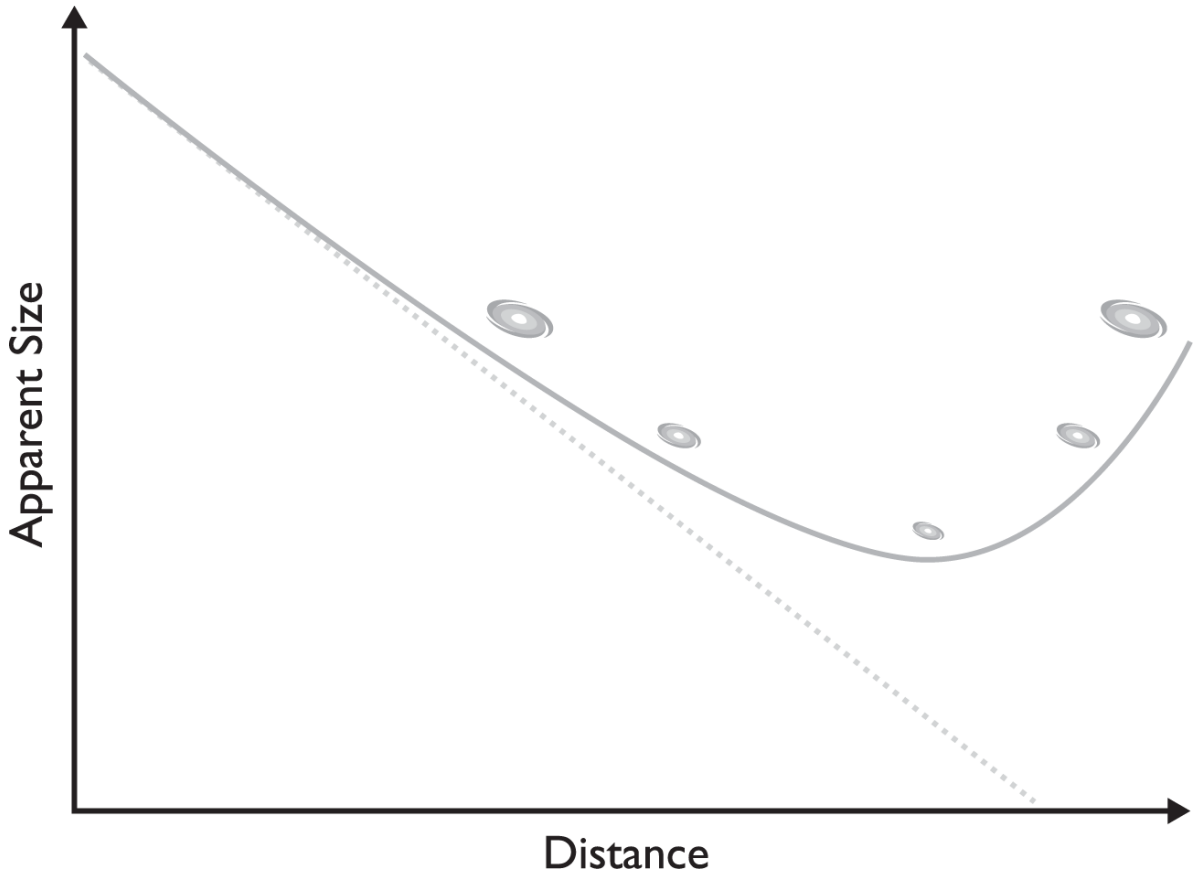


Figure 12: The apparent size of distant galaxies (assuming the same physical size) as a function of distance from us. Out to a certain distance, a faraway galaxy will appear to be smaller, but at some point, this turns around, and a more distant galaxy will look bigger in the sky. The dotted line indicates how the apparent size would relate to distance in a static universe.

The reason for this reversal is related to the reason we can see things that are currently moving away from us faster than light. In the past, when the light was emitted, they were closer. So close, in fact, that they covered more of the sky. Even though they're much farther away now, the "snapshot" they've sent us has been traveling all that time, and is just reaching us now, showing us the ghostly image of a much closer thing. And the farther back in time you go, the smaller the universe was. So beyond a certain point, the balance between "the universe was smaller in the past" and "light takes a certain amount of time to get here" is such that a galaxy that is more distant than another galaxy *now* might have actually been *closer* when its light was emitted.

Look, I warned you it would be weird.

Anyway, if this is all deeply confusing and mind-boggling, that's totally okay and normal. Maybe try drawing some sketches on napkins, and then